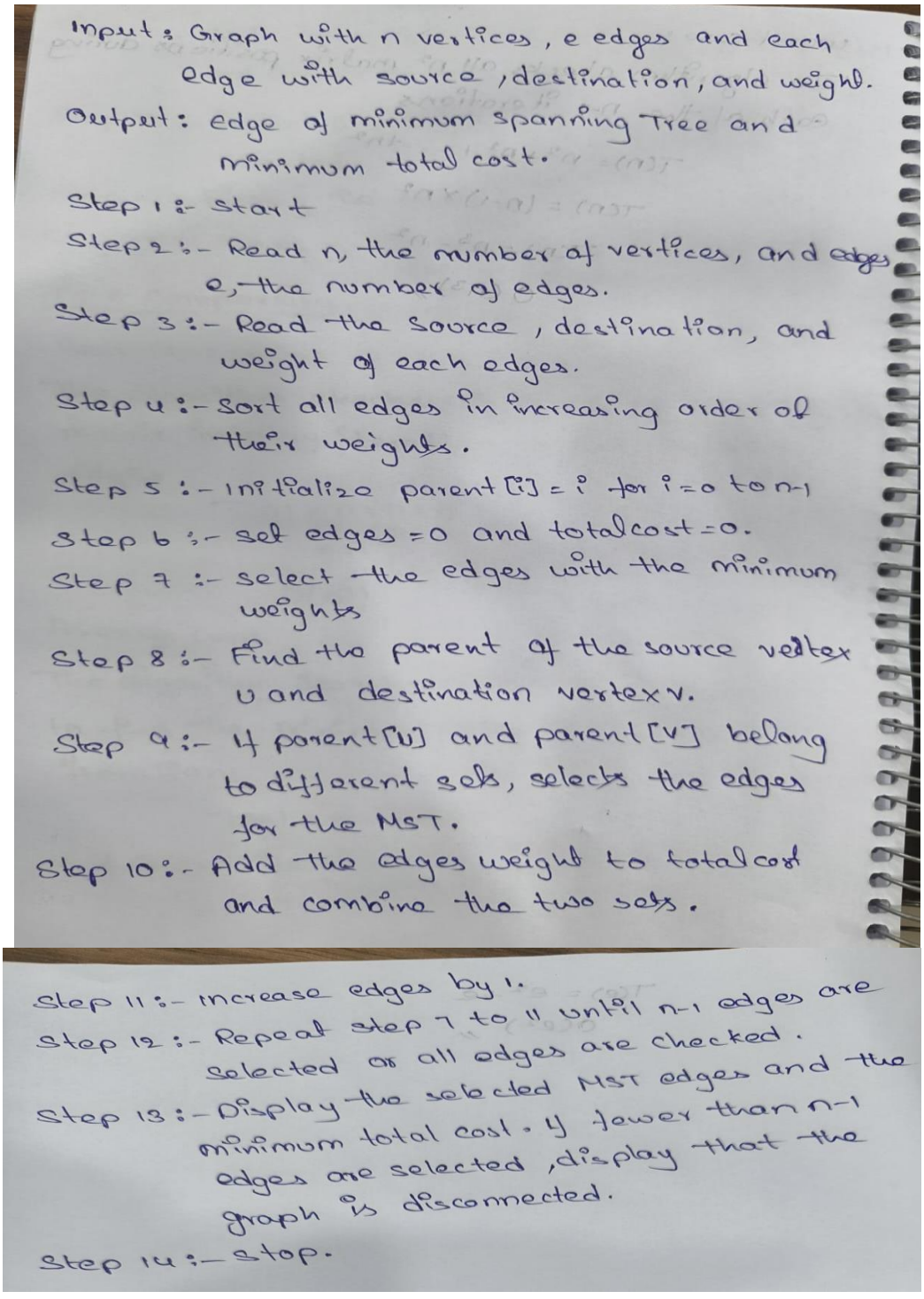
Practical- 10

Aim: To Implement Kruskal's algorithm.

Kruskal algorithm

Algorithm:



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Program:

```
#include <iostream>
using namespace std;

class Graph
{
public:
   int u[20], v[20], w[20];
   int parent[10];
   int n, e;

   void getInput()
   {
      cout << "Enter number of vertices: ";
      cin >> n;

      cout << "Enter number of edges: ";
      cin >> e;

      cout << "Enter source, destination and weight:\n";

      for(int i = 0; i < e; i++)
         cin >> u[i] >> v[i] >> w[i];
   }

   int find(int x)
   {
      while(parent[x] != x)
         x = parent[x];

      return x;
   }

   void kruskal()
   {
      int temp, totalCost = 0, edges = 0;
      int a, b;

      for(int i = 0; i < n; i++)
         parent[i] = i;

      for(int i = 0; i < e - 1; i++)
      {
         for(int j = 0; j < e - i - 1; j++)
         {
            if(w[j] > w[j + 1])
            {
               temp = w[j];
```

```
                w[j] = w[j + 1];
                w[j + 1] = temp;

                temp = u[j];
                u[j] = u[j + 1];
                u[j + 1] = temp;

                temp = v[j];
                v[j] = v[j + 1];
                v[j + 1] = temp;
            }
        }
    }

    cout << "\nEdges in Minimum Spanning Tree:\n";

    for(int i = 0; i < e && edges < n - 1; i++)
    {
        a = find(u[i]);
        b = find(v[i]);

        if(a != b)
        {
            cout << u[i] << " - " << v[i]
                 << " = " << w[i] << endl;

            parent[a] = b;
            totalCost += w[i];
            edges++;
        }
    }

    if(edges != n - 1)
        cout << "Graph is disconnected\n";
    else
        cout << "\nMinimum Cost = " << totalCost << endl;
  }
};

int main()
{
  Graph g;

  g.getInput();
  g.kruskal();

  return 0;
}
```

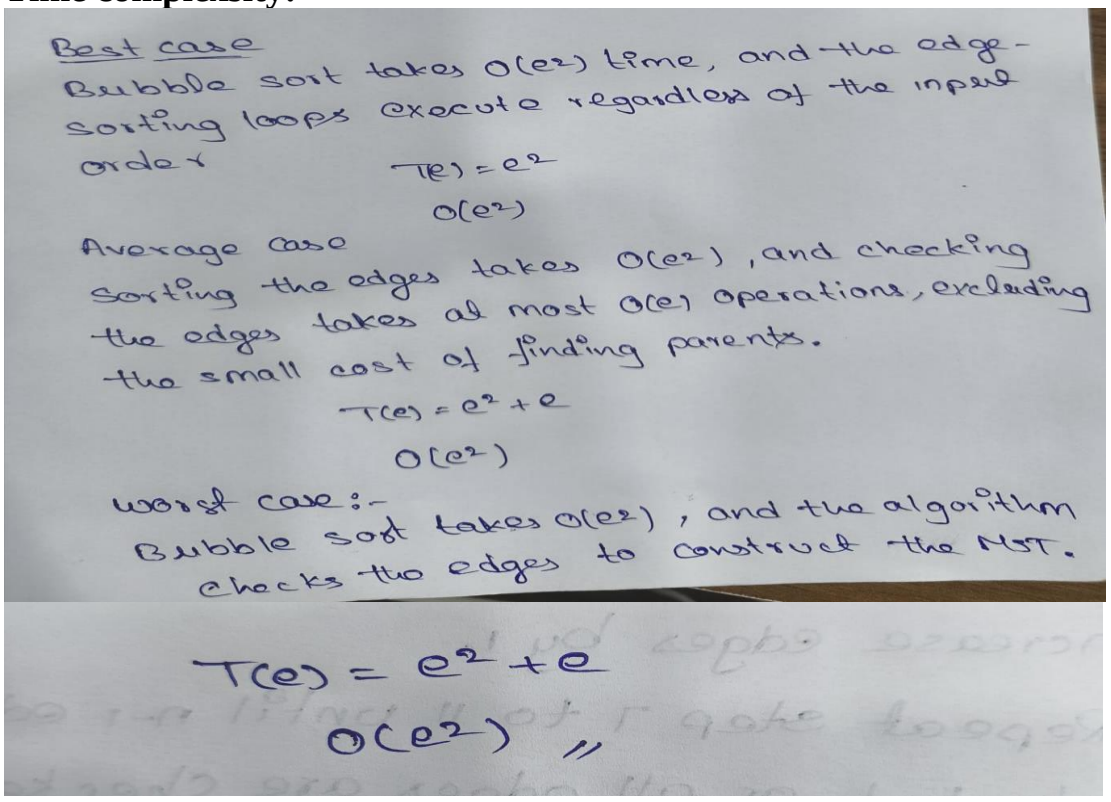
Output:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter source, destination and weight:
0
1
2
0
2
4
1
2
1
1
3
5
2
3
3

Edges in Minimum Spanning Tree:
1 - 2 = 1
0 - 1 = 2
2 - 3 = 3

Minimum Cost = 6
```

Time complexsity:



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Conclusion:

Thus, Kruskal's algorithm was successfully implemented to find the Minimum Spanning Tree (MST) of a connected, weighted graph. It selects the edges with the lowest weights while avoiding cycles until all vertices are connected. Kruskal's algorithm is useful in network design, road construction, and communication systems to minimize the total cost.

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